

Railway applications — Aerodynamics —

Part 1: Symbols and units

The European Standard EN 14067-1:2003 has the status of a
British Standard

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Confirmed September 2011

National foreword

This British Standard is the official English language version of EN 14067-1:2003.

The UK participation in its preparation was entrusted by Technical Committee RAE/1, Railway applications, to Subcommittee RAE/1/-/4, Aerodynamics, which has the responsibility to:

- aid enquirers to understand the text;
- present to the responsible international/European committee any enquiries on the interpretation, or proposals for change, and keep the UK interests informed;
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Summary of pages

This document comprises a front cover, an inside front cover, the EN title page, pages 2 to 10, an inside back cover and a back cover.

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Amendments issued since publication

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English version

Railway applications - Aerodynamics - Part 1: Symbols and units

Applications ferroviaires - Aérodynamique - Partie 1:
Symboles et unités

Bahnanwendungen - Aerodynamik - Teil 1: Formelzeichen
und Einheiten

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EUROPÄISCHES KOMITEE FÜR NORMUNG

Management Centre: rue de Stassart, 36 B-1050 Brussels

Contents

Foreword3

1 Scope3

2 Terms, definitions, symbols and abbreviations4

Foreword

This document EN 14067-1:2003 has been prepared by Technical Committee CEN /TC 256, "Railway applications", the secretariat of which is held by DIN.

This European Standard shall be given the status of a national standard, either by publication of an identical text or by endorsement, at the latest by October 2003, and conflicting national standards shall be withdrawn at the latest by October 2003.

This European Standard is part of the series "Railway applications — Aerodynamics" which consists of the following parts:

- Part 1: Symbols and units
- Part 2: Aerodynamics on open track
- Part 3: Aerodynamics in tunnels
- Part 4: Requirements and test procedures for aerodynamics on open track¹⁾
- Part 5: Requirements and test procedures for aerodynamics in tunnels¹⁾

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1 Scope

This European Standard applies to aerodynamics for railway applications.

It defines symbols and units used in formulae and calculations in the field of aerodynamics. The definitions given in this European Standard explain the symbols and classify the units.

Further to usual symbols, this document contains symbols which are used for calculations. It should be noted, however, that these symbols may not have the same significance in a different field of application.

Additional symbols should be defined according to the application.

1) in preparation

2 Terms, definitions, symbols and abbreviations

All parameters are expressed as SI basic units and units derived from SI basic units.

Table/ Tableau/ Tabelle 1

Symbol Symboles Symbol	Unit Unité Einheit	Significance Signification Bedeutung	Explanation Explication Erläuterung	Remark Remarque Bemerkung
A	m^2	surface area section transversale Oberfläche		
a	m/s^2	acceleration accélération Beschleunigung		
B	-	train/tunnel blockage ratio rapport de blocage Versperrungsmaß		$B = \frac{S_{tr}}{S_{tu}}$
b	m	width of vehicle largeur du véhicule Fahrzeugbreite		
c	m/s	speed of sound vitesse du son Schallgeschwindigkeit		
C_1	N	rolling resistance in the resistance to motion formula résistance au roulement (dans la formule de résistance à l'avancement) Rollwiderstand in der Laufwiderstandsformel		
$C_2 v_{tr}$	N	momentum resistance in the resistance to motion formula résistance due aux systèmes de ventilation (dans la formule de résistance à l'avancement) Impuls widerstand in der Laufwiderstandsformel		$C_2 = Qp,$
$C_3 v_{tr}^2$	N	aerodynamic drag in the resistance to motion formula traînée aérodynamique (dans la formule de résistance à l'avancement) Aerodynamischer Widerstand in der Laufwiderstandsformel		
C_B	-	gust coefficient coefficient de rafale Böenkoeffizient		$C_B = \frac{v_B}{v_w}$
C_F	-	coefficient of aerodynamic force coefficient de la force aérodynamique Koeffizient der Luftkraft		$C_F = \frac{2 F}{S \rho v_{tr}^2}$
C_f	-	coefficient of skin-friction drag coefficient de frottement aérodynamique Koeffizient des Luftreibungswiderstandes		$C_f = \frac{R_f}{q A}$
C_p	-	coefficient of pressure amplitude coefficient de l'amplitude de pression Koeffizient der Druckamplitude		$C_p = \frac{p - p_o}{q}$
ΔC_p	-	pressure coefficient coefficient de pression Druckkoeffizient		$\Delta C_p = \frac{2 (p_{\max} - p_{\min})}{\rho v_{tr}^2}$

Table/ Tableau/ Tabelle 1 (continued/ suite/ fortgesetzt)

Symbol Symboles Symbol	Unit Unité Einheit	Significance Signification Bedeutung	Explanation Explication Erläuterung	Remark Remarque Bemerkung
C_{Fx} C_{Fy} C_{Fz}	-	coefficients of aerodynamic forces F_x, F_y, F_z coefficients de force aérodynamique F_x, F_y, F_z Koeffizienten der Luftkräfte F_x, F_y, F_z		$C_{F_x} = \frac{F_x}{q S}; C_{F_y} = \frac{F_y}{q S}; C_{F_z} = \frac{F_z}{q S}$
C_{Mx} C_{My} C_{Mz}	-	coefficients of the moments M_x, M_y, M_z coefficients de moment M_x, M_y, M_z Koeffizienten der Momente M_x, M_y, M_z		$C_{M_x} = \frac{M_x}{q S D_h}; C_{M_y} = \frac{M_y}{q S D_h}$ $C_{M_z} = \frac{M_z}{q S D_h}$
C_x	-	coefficient of the aerodynamic drag for the whole train coefficient de traînée aérodynamique pour le train entier Koeffizient des aerodynamischen Widerstands für den gesamten Zug		
C_{x1}	-	coefficient of the aerodynamic drag of the leading vehicle coefficient de traînée aérodynamique du véhicule de tête Koeffizient des aerodynamischen Widerstands des führenden Fahrzeugs		
C_{x2}	-	coefficient of the aerodynamic drag of the train without leading vehicle and tail coefficient de traînée aérodynamique de la partie du train hors véhicules de tête et de queue Koeffizient des aerodynamischen Widerstands des Zuges ohne führendes Fahrzeug und ohne Heck		
C_{x3}	-	coefficient of the aerodynamic drag of the tail vehicle coefficient de traînée aérodynamique du véhicule de queue Koeffizient des aerodynamischen Widerstands des Endfahrzeugs		
C_p	-	specific heat at constant pressure chaleur spécifique à pression constante Spezifische Wärme bei konstantem Druck		
C_v	-	specific heat at constant volume chaleur spécifique à volume constant Spezifische Wärme bei konstantem Volumen		
D	m	diameter diamètre Durchmesser		$D = 2 r$
D_h	m	hydraulic diameter (European definition) diamètre hydraulique (définition européenne) hydraulischer Durchmesser (europäische Definition)		$D_h = \frac{4 S}{Pe}$
d	m	characteristic dimension longueur caractéristique charakteristische Länge		
F	N	aerodynamic force force aérodynamique Luftkraft		

Table/ Tableau/ Tabelle 1 (continued/ suite/ fortgesetzt)

Symbol Symboles Symbol	Unit Unité Einheit	Significance Signification Bedeutung	Explanation Explication Erläuterung	Remark Remarque Bemerkung
F_x F_y F_z	N	aerodynamic forces in the directions of coordinates composantes de la force aérodynamique dans le repère cartésien Luftkräfte in Richtung der Koordinatenachsen	Figure/ Figure/ Bild 1	
f	Hz	frequency fréquence Frequenz		
g	m/s ²	acceleration due to gravity accélération due à la gravité Erdbeschleunigung		
h	m	height hauteur Höhe		
i	°/°	gradient of track gradient de la voie (pente, rampe) Steigung, Neigung der Strecke		
k	-	pressure loss coefficient coefficient de perte de charge Druckverlustkoeffizient		
K_s	m	equivalent sand roughness rugosité équivalente du sable Äquivalente Sandrauigkeit		
L_{loco}	m	length of the locomotive longueur de la locomotive Länge der Lokomotive		
L_p	dB	sound pressure level niveau de pression acoustique Schalldruckpegel		
L_n	m	length of the non-cylindrical part of the leading car, (nose length) longueur de la partie à section non constante du véhicule de tête, (longueur de la tête) Länge des nicht-zylindrischen Teils des führenden Fahrzeugs, (Kopflänge)		
L_{tr}	m	length of train longueur du train Zuglänge		
L_{tu}	m	length of tunnel longueur du tunnel Tunnellänge		
Ma	-	Mach number nombre de Mach Machzahl		
M_x	Nm	aerodynamic rolling moment moment de roulis aérodynamique Aerodynamisches Rollmoment	Figure/ Figure/ Bild 1	
M_y	Nm	aerodynamic pitching moment moment de tangage aérodynamique Aerodynamisches Nickmoment	Figure/ Figure/ Bild 1	

Table/ Tableau/ Tabelle 1 (continued/ suite/ fortgesetzt)

Symbol Symboles Symbol	Unit Unité Einheit	Significance Signification Bedeutung	Explanation Explication Erläuterung	Remark Remarque Bemerkung
M_z	Nm	aerodynamic yawing moment moment de lacet aerodynamisches Giermoment	Figure/ Figure/ Bild 1	
m	kg	mass masse Masse		
N	-	number of trailer cars nombre de véhicules Anzahl der Fahrzeuge		
P	W	power puissance Leistung		
Pe	m	perimeter périmètre Umfang		
p	Pa	pressure pression Druck		
p_o	Pa	reference static pressure pression statique de référence statischer Bezugsdruck		
$p_{\max}$	Pa	maximum pressure pression maximale Höchstdruck		
$p_{\min}$	Pa	minimum pressure pression minimale Mindestdruck		
p_t	Pa	total pressure pression totale Gesamtdruck		
Q	m ³ /s	volume flow rate débit volumétrique Volumenstrom		
Q_m	kg/s	mass flow rate débit massique Durchflußmenge		
q	Pa	dynamic pressure pression dynamique Staudruck (dynamischer Druck)		$q = \frac{\rho U^2}{2}$
R	N	running resistance résistance à l'avancement Laufwiderstand		
R_a	N	aerodynamic drag traînée aérodynamique aerodynamischer Widerstand (Luftwiderstand)		$R_a = -F_x$
Re	-	Reynolds number nombre de Reynolds Reynoldszahl		$Re = \frac{U d}{\nu}$

Table/ Tableau/ Tabelle 1 (continued/ suite/ fortgesetzt)

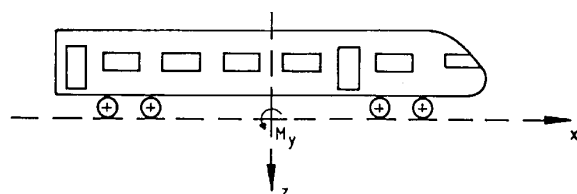
Symbol Symboles Symbol	Unit Unité Einheit	Significance Signification Bedeutung	Explanation Explication Erläuterung	Remark Remarque Bemerkung
R_f	N	frictional drag résistance aérodynamique due au frottement (Luft-) Reibungswiderstand		$R_f = C_f q A$
R_g	N	grade resistance résistance à l'avancement due au profil Steigungswiderstand		$R_g = m g i$
r	m	radius rayon Radius		$r = \frac{D}{2}$
S	m ²	cross-sectional area, characteristic area section transversale, surface caractéristique Querschnittsfläche, charakteristische Fläche		
S_{tr}	m ²	cross-sectional area of the train section transversale du train Querschnittsfläche des Zuges		
S_{tu}	m ²	cross-sectional area of tunnel section transversale du tunnel Querschnittsfläche des Tunnels		
s	m	track gauge écartement de la voie Spurweite		
T	K	absolute temperature température absolue Absolute Temperatur		
T_f	-	tunnel factor coefficient de tunnel Tunnelfaktor		$T_f = \frac{\text{aerodynamic drag in the tunnel}}{\text{aerodynamic drag in the open air}}$ $T_f = \frac{\text{trainée aérodynamique en tunnel}}{\text{trainée aérodynamique à l'air libre}}$ $T_f = \frac{\text{aerodynamischer Widerstand im Tunnel}}{\text{aerodynamischer Widerstand auf freier Strecke}}$
t	s	time temps Zeit		
Δt	s	time interval durée Zeitintervall		
U	m/s	flow speed vitesse de l'écoulement Strömungsgeschwindigkeit		
v_a	m/s	air speed relative to the train vitesse relative de l'air vue par le train relative Anströmgeschwindigkeit	Figure/ Figure/ Bild 1	
v_B	m/s	gust speed vitesse de la rafale Böengeschwindigkeit		

Table/ Tableau/ Tabelle 1 (continued/ suite/ fortgesetzt)

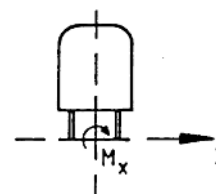
Symbol Symboles Symbol	Unit Unité Einheit	Significance Signification Bedeutung	Explanation Explication Erläuterung	Remark Remarque Bemerkung
v_{tr}	m/s	train speed vitesse du train Zuggeschwindigkeit	Figure/ Figure/ Bild 1	km/h may also be used – this shall be indicated km/h peuvent aussi être utilisés – doit être indiqué km/h dürfen auch verwendet werden – ist anzugeben
v_w	m/s	wind speed vitesse du vent Windgeschwindigkeit	Figure/ Figure/ Bild 1	
x y z	- - -	cartesian coordinates coordonnées cartésiennes Kartesische Koordinaten	Figure/ Figure/ Bild 1	x : Origin is halfway between the axles of the inner wheels l'origine est au milieu des essieux en partant de la face interne des roues Ursprung ist die Mitte zwischen den inneren Radachsen y : Origin is track centre line l'origine est l'axe de la voie Ursprung ist Gleismitte z : Origin is top of rail l'origine est la face supérieure du rail Ursprung ist Schienenoberkante
Y	m	distance from track centre line distance par rapport à l'axe de la voie Abstand von der Gleismitte		
Y_s	m	distance from coach side distance par rapport à la paroi du véhicule Abstand von der Fahrzeugseitenwand		
θ	°C	temperature température Temperatur		
β	°	yaw angle angle de lacet Schiebewinkel	Figure/ Figure/ Bild 1	
β_w	°	wind angle relative to track angle formé par la direction du vent et la voie Winkel zwischen Wind- und Gleisrichtung	Figure/ Figure/ Bild 1	
γ	-	slenderness ratio finesse aérodynamique Schlankheitsgrad		$\gamma = \frac{L_n}{D_h}$
δ	m	boundary-layer thickness épaisseur de la couche limite Grenzschichtdicke		
κ	-	ratio of specific heats rapport des chaleurs spécifiques Verhältnis der spezifischen Wärmen		$\kappa = c_p/c_v$

Table/ Tableau/ Tabelle 1 (continued/ suite/ fortgesetzt)

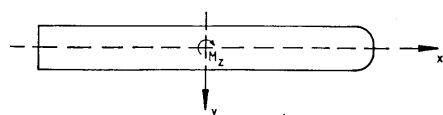
Symbol Symboles Symbol	Unit Unité Einheit	Significance Signification Bedeutung	Explanation Explication Erläuterung	Remark Remarque Bemerkung
ν	m^2/s	kinematic viscosity viscosité cinématique Kinematische Viskosität		
ρ	kg/m^3	density densité Dichte		
ζ_h	-	head loss coefficient coefficient de perte à la tête Kopfverlustbeiwert		
ζ_k	-	coefficient, depends on the shape of the train head coefficient, dépend de la forme de la tête du train Beiwert, von der Kopfform abhängig		



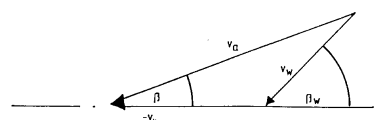
Key: a) view from side
vue latérale
Seitenansicht



Key: b) view from behind
vue arrière
Rückansicht



Key: c) view from top
vue de dessus
Draufsicht



Key: d) speed vector diagram
triangle de vitesse
Geschwindigkeitsdreieck

Figure/ Figure/ Bild 1

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